AI Study Assistant Using Ollama and Streamlit

Internship Project Report

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1. Project Objective

The objective of this project is to build a simple AI question-answering application using Ollama as the local AI model runtime and Streamlit as the user interface. The application allows a user to enter a question and receive a response generated by a locally running Llama 3.2 model.

2. Technologies Used

Technology	Purpose
Python 3.13.13	Programming language used to build the application.
Ollama 0.34.4	Runs the AI model locally on the computer.
Llama 3.2	Local language model used to generate answers.
Streamlit 1.64.0	Provides the web-based user interface.
Ollama Python client 0.6.2	Connects the Python application to Ollama.

3. What is Ollama?

Ollama is used in this project to run the selected language model locally. The project uses Llama 3.2 through Ollama, so prompts can be sent to the local model from the Python application without requiring a separate hosted AI service.

4. What is Streamlit?

Streamlit is a Python framework used to create a simple interactive web interface. In this project, it provides the title, question input area, Generate Answer button, loading indicator, and AI Response section.

5. How the Application Works

Step	Process
1	User enters a question in the Streamlit text area.
2	The user clicks the Generate Answer button.
3	The Python application sends the prompt to Llama 3.2 through the Ollama client.
4	The local model generates a response.
5	Streamlit displays the generated response under AI Response.

6. Testing Performed

No.	Prompt Tested
1	What is artificial intelligence? Explain it in simple words.
2	What is the difference between Python and Java?
3	Explain the concept of a database and give two examples.
4	Explain machine learning and give three real-world applications.
5	What are the main benefits of renewable energy?

7. Screenshots of the Working Application

Figure 1: Streamlit application output

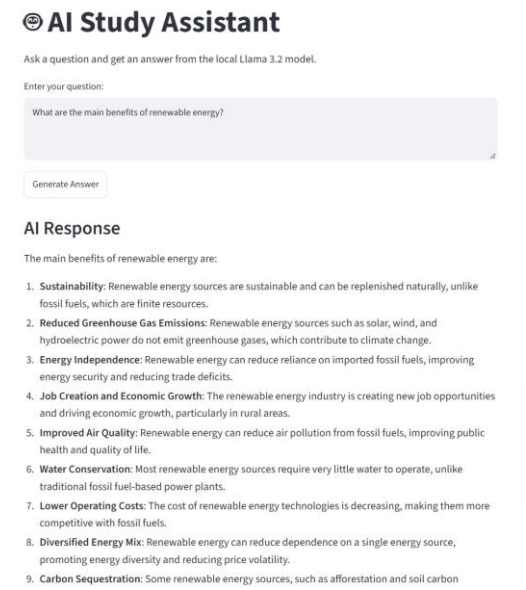


Figure 2: Streamlit application output

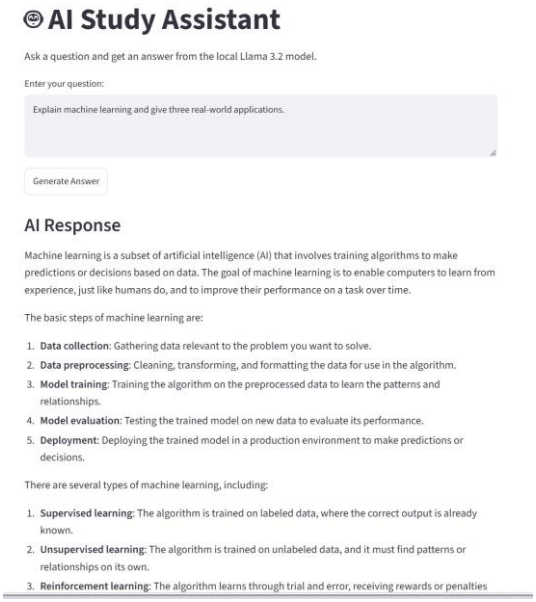
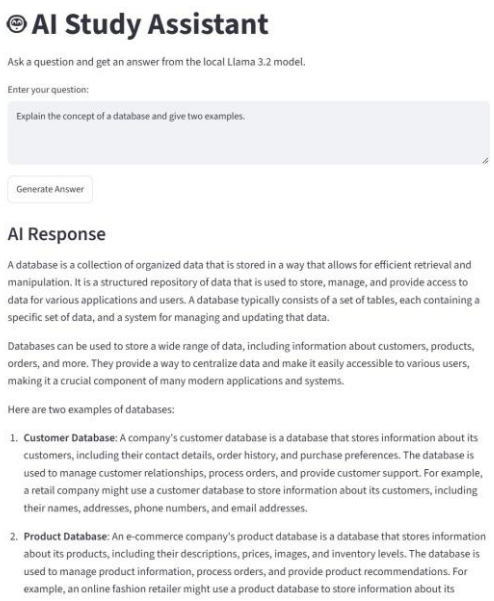


Figure 3: Streamlit application output



8. Learning Outcomes

Through this project, I learned how to set up and run a local large language model using Ollama, communicate with the model through its Python client, and build an interactive user interface using Streamlit. I also learned how to create a Python virtual environment, install project dependencies, test an AI application with different prompts, and organize the files required for submission.

9. Conclusion

The project successfully demonstrates a simple local AI question-answering system. A user can enter a prompt in the Streamlit interface, the prompt is processed by the locally running Llama 3.2 model through Ollama, and the generated response is displayed back in the application.